

UCS301 Data Structures
Lab Assignment 4: Queues
(Week 4)

- 1) Develop a menu driven program demonstrating the following operations on simple Queues: enqueue(), dequeue(), isEmpty(), isFull(), display(), and peek().
- 2) Develop a menu driven program demonstrating the following operations on Circular Queues: enqueue(), dequeue(), isEmpty(), isFull(), display(), and peek().
- 3) Write a program to interleave the first half of the queue with the second half.
Sample I/P: 4 7 11 20 5 9 Sample O/P: 4 20 7 5 11 9
- 4) Write a program to find the first non-repeating character in a string using a Queue. Sample I/P: a a b c Sample O/P: a -1 b b
- 5) Write a program to implement a stack using (a) Two queues and (b) One Queue.